

Research Analysis Report

Research Topic:

Total Papers: 0

Generated By:

Generated At:

Executive Summary

Literature Review

Methodology Comparison

Research Gap Analysis

Total Papers: 5

Research Areas

- Agentic AI

Common Keywords

- ability
- adaptability
- agent
- agentic
- analysis
- application
- artificial
- automation
- autonomous
- autonomy
- between
- beyond
- capable
- challenge

- complex
- conceptual
- critically
- decision-making
- designed
- discovery
- distinguishe
- emerged
- exhibiting
- focusing
- frontier
- generative
- goal-driven
- goals
- goes
- hold
- human
- important
- integration
- intelligence
- intervention
- language
- large
- llms
- mapping
- marks
- minimal
- model
- near-human
- offering
- often
- opportunity
- performance
- planning
- praised
- pursue
- range
- reasoning

- recently
- refer
- review
- scientific
- scientist
- structured
- system
- tasks
- taxonomy
- traditional
- transforming
- unlike
- valued
- wide

Future Work

- Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration
- Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI
- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts

Research Area Distribution

Agentic AI: 5

Keyword Frequency

agentic: 4
autonomous: 3
system: 3
traditional: 2
reasoning: 2
artificial: 1
intelligence: 1
refer: 1
designed: 1
pursue: 1
complex: 1
goals: 1

minimal: 1
human: 1
intervention: 1
unlike: 1
review: 1
critically: 1
distinguish: 1
between: 1
agent: 1
offering: 1
structured: 1
conceptual: 1
taxonomy: 1
application: 1
mapping: 1
analysis: 1
opportunity: 1
challenge: 1
large: 1
language: 1
model: 1
llms: 1
often: 1
praised: 1
exhibiting: 1
near-human: 1
performance: 1
wide: 1
range: 1
tasks: 1
valued: 1
ability: 1
hold: 1
recently: 1
emerged: 1
important: 1
goes: 1
beyond: 1

generative: 1
focusing: 1
autonomy: 1
adaptability: 1
goal-driven: 1
integration: 1
scientific: 1
discovery: 1
marks: 1
frontier: 1
automation: 1
capable: 1
planning: 1
decision-making: 1
transforming: 1
scientist: 1

Research Trends

- agentic
- autonomous
- system
- traditional
- reasoning
- artificial
- intelligence
- refer
- designed
- pursue
- complex
- goals
- minimal
- human
- intervention

Gap Categories

datasets: []

models: ['Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI']

evaluation: []

applications: []

general: ['Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration', 'We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts']

Emerging Topics

- agentic
- autonomous
- system
- traditional
- reasoning

Recommendations

- Explore autonomous multi-agent systems.
- Improve reasoning capabilities using LLM planning.

Summary

dominant_area: Agentic AI

top_trend: agentic

future_work_items: 3

Experiment Plan

Research Areas

- Agentic AI

Recommended Datasets

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- OpenAI Evals

Baseline Models

None

Evaluation Metrics

- Accuracy

- Precision
- Recall
- F1-Score
- ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

- Train-Test Split
- K-Fold Cross Validation
- Hyperparameter Tuning
- Statistical Significance Testing

Experimental Workflow

- Collect Dataset
- Preprocess Data
- Train Baseline Models
- Train Proposed Model
- Evaluate Performance
- Compare Results
- Analyze Errors
- Draw Conclusions

Report Summary

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.